

CORE PURE / DIFFERENTIAL EQUATIONS

Modelling with 1st Order Differential Equations

Worked solutions

14 questions · 220 marks

Questions

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1. A laboratory mixing vessel initially contains 4 litres of solution in which 4 mg of dye is dissolved.

A wash solution enters the vessel at a constant rate of 300 ml per minute.

At the same time, well-mixed liquid leaves the vessel at a constant rate of 200 ml per minute.

The concentration of dye in the liquid entering the vessel, at time t minutes after pumping begins, is modelled by $2e^{-0.1t}$ mg per litre.

Let m be the amount of dye, in milligrams, in the vessel at time t minutes.

- (a) Show that the volume of liquid in the vessel is $(4 + 0.1t)$ litres, and hence show that m satisfies

$$\frac{dm}{dt} = 0.6e^{-0.1t} - \frac{2m}{40 + t} \quad [4]$$

- (b) Solve this differential equation, given that $m = 4$ when $t = 0$, to find an expression for m in terms of t [8]

- (c) After 20 minutes, the concentration of dye in the vessel was measured as 0.82 mg per litre. Assess the model in light of this information, giving a reason for your answer [2]

Solution

- (a) The liquid enters at 300 ml/min and leaves at 200 ml/min, so the net increase in volume is

$$300 - 200 = 100 \text{ ml/min} = 0.1 \text{ L/min}$$

Hence, starting from 4 L, the volume after t minutes is

$$V(t) = 4 + 0.1t$$

Now form a rate equation for the dye.

The dye enters with concentration $2e^{-0.1t}$ mg/L and flow rate 0.3 L/min, so the rate of dye entering is

$$0.3 \times 2e^{-0.1t} = 0.6e^{-0.1t} \text{ mg/min}$$

The concentration of dye in the vessel at time t is

$$\frac{m}{4 + 0.1t} \text{ mg/L}$$

Since liquid leaves at 0.2 L/min, the rate of dye leaving is

$$0.2 \left(\frac{m}{4 + 0.1t} \right)$$

Simplifying,

$$0.2 \left(\frac{m}{4 + 0.1t} \right) = \frac{0.2m}{4 + 0.1t} = \frac{2m}{40 + t}$$

Therefore,

$$\begin{aligned} \frac{dm}{dt} &= \text{rate in} - \text{rate out} \\ &= 0.6e^{-0.1t} - \frac{2m}{40 + t} \end{aligned}$$

So

$$V(t) = 4 + 0.1t \text{ litres,} \quad \frac{dm}{dt} = 0.6e^{-0.1t} - \frac{2m}{40 + t}$$

(b) We solve

$$\frac{dm}{dt} + \frac{2m}{40+t} = 0.6e^{-0.1t}$$

This is a linear differential equation.

The integrating factor is

$$\begin{aligned}\mu(t) &= \exp\left(\int \frac{2}{40+t} dt\right) \\ &= \exp(2 \ln(40+t)) \\ &= (40+t)^2\end{aligned}$$

Multiplying the differential equation by $(40+t)^2$,

$$(40+t)^2 \frac{dm}{dt} + (40+t)^2 \frac{2m}{40+t} = 0.6(40+t)^2 e^{-0.1t}$$

So

$$\frac{d}{dt} ((40+t)^2 m) = 0.6(40+t)^2 e^{-0.1t}$$

Integrating both sides,

$$(40+t)^2 m = \int 0.6(40+t)^2 e^{-0.1t} dt$$

Expand $(40+t)^2$:

$$(40+t)^2 = t^2 + 80t + 1600$$

So

$$(40+t)^2 m = \int 0.6(t^2 + 80t + 1600)e^{-0.1t} dt$$

Let

$$I = \int 0.6(t^2 + 80t + 1600)e^{-0.1t} dt$$

and use integration by parts with

$$u = t^2 + 80t + 1600, \quad \frac{dv}{dt} = 0.6e^{-0.1t}$$

Then

$$\frac{du}{dt} = 2t + 80, \quad v = \int 0.6e^{-0.1t} dt = -6e^{-0.1t}$$

So

$$\begin{aligned}I &= uv - \int v \frac{du}{dt} dt \\ &= -6(t^2 + 80t + 1600)e^{-0.1t} + 6 \int (2t + 80)e^{-0.1t} dt \\ &= -6(t^2 + 80t + 1600)e^{-0.1t} + 12 \int (t + 40)e^{-0.1t} dt\end{aligned}$$

Now let

$$J = \int (t + 40)e^{-0.1t} dt$$

Again integrating by parts, take

$$u = t + 40, \quad \frac{dv}{dt} = e^{-0.1t}$$

Then

$$\frac{du}{dt} = 1, \quad v = \int e^{-0.1t} dt = -10e^{-0.1t}$$

Hence

$$\begin{aligned}
 J &= uv - \int v \frac{du}{dt} dt \\
 &= -10(t+40)e^{-0.1t} + 10 \int e^{-0.1t} dt \\
 &= -10(t+40)e^{-0.1t} - 100e^{-0.1t} + K \\
 &= -10(t+50)e^{-0.1t} + K
 \end{aligned}$$

Substituting this into the expression for I ,

$$\begin{aligned}
 I &= -6(t^2 + 80t + 1600)e^{-0.1t} + 12(-10(t+50)e^{-0.1t}) + K \\
 &= -6e^{-0.1t}(t^2 + 80t + 1600 + 20t + 1000) + K \\
 &= -6e^{-0.1t}(t^2 + 100t + 2600) + K
 \end{aligned}$$

Therefore,

$$\int 0.6(t^2 + 80t + 1600)e^{-0.1t} dt = -6e^{-0.1t}(t^2 + 100t + 2600) + K$$

where K is a constant.

So

$$(40+t)^2 m = -6e^{-0.1t}(t^2 + 100t + 2600) + K$$

Now use the initial condition $m = 4$ when $t = 0$:

$$\begin{aligned}
 (40)^2(4) &= -6e^0(0^2 + 100(0) + 2600) + K \\
 6400 &= -6(2600) + K \\
 6400 &= -15600 + K
 \end{aligned}$$

Thus

$$K = 22000$$

Hence

$$(40+t)^2 m = 22000 - 6e^{-0.1t}(t^2 + 100t + 2600)$$

and so

$$m = \frac{22000 - 6e^{-0.1t}(t^2 + 100t + 2600)}{(40+t)^2}$$

Therefore, the amount of dye is

$$m(t) = \frac{22000 - 6e^{-0.1t}(t^2 + 100t + 2600)}{(40+t)^2} \text{ mg}$$

(c) After 20 minutes, the model gives volume

$$V(20) = 4 + 0.1(20) = 6 \text{ L}$$

Using the expression for m ,

$$\begin{aligned}
 m(20) &= \frac{22000 - 6e^{-2}(20^2 + 100(20) + 2600)}{(40+20)^2} \\
 &= \frac{22000 - 6e^{-2}(400 + 2000 + 2600)}{60^2} \\
 &= \frac{22000 - 30000e^{-2}}{3600} \\
 &= \frac{55}{9} - \frac{25}{3}e^{-2}
 \end{aligned}$$

So the predicted concentration is

$$\begin{aligned}\frac{m(20)}{V(20)} &= \frac{1}{6} \left(\frac{55}{9} - \frac{25}{3} e^{-2} \right) \\ &= \frac{55}{54} - \frac{25}{18} e^{-2} \\ &\approx 0.831 \text{ mg/L}\end{aligned}$$

The measured concentration was 0.82 mg/L, which is very close to 0.831 mg/L.

So the model appears reasonable, since its prediction is consistent with the observed value.

2. A particle B , of mass 2 kg, moves in a straight line and has velocity $v \text{ m s}^{-1}$.

At time t seconds, where $t \geq 0$, a variable force of $\frac{12t - 4v}{1 + t}$ N acts on B .

There are no other forces acting on B . Initially, when $t = 0$, B has velocity -5 m s^{-1} .

The motion of B can be modelled by the differential equation

$$\frac{dv}{dt} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t .

- (a) Find the functions $P(t)$ and $Q(t)$ [2]

- (b) Using an integrating factor, determine the first time at which B is stationary according to the model [8]

Solution

- (a) Since the mass of B is 2 kg and the only force acting is

$$\frac{12t - 4v}{1 + t}$$

Newton's second law gives

$$2 \frac{dv}{dt} = \frac{12t - 4v}{1 + t}$$

Rearranging into the form

$$\frac{dv}{dt} + P(t)v = Q(t)$$

gives

$$\begin{aligned} \frac{dv}{dt} &= \frac{12t - 4v}{2(1 + t)} \\ &= \frac{6t - 2v}{1 + t} \\ \frac{dv}{dt} + \frac{2}{1 + t}v &= \frac{6t}{1 + t} \end{aligned}$$

Hence

$$P(t) = \frac{2}{1 + t} \quad \text{and} \quad Q(t) = \frac{6t}{1 + t}$$

- (b) From part (a), the differential equation is

$$\frac{dv}{dt} + \frac{2}{1 + t}v = \frac{6t}{1 + t}$$

The integrating factor is

$$\begin{aligned} \mu(t) &= e^{\int \frac{2}{1+t} dt} \\ &= e^{2 \ln(1+t)} \\ &= (1 + t)^2 \end{aligned}$$

since $t \geq 0$, so $1 + t > 0$.

Multiplying the differential equation by $(1 + t)^2$ gives

$$\begin{aligned} (1 + t)^2 \frac{dv}{dt} + \frac{2(1 + t)^2}{1 + t}v &= \frac{6t(1 + t)^2}{1 + t} \\ (1 + t)^2 \frac{dv}{dt} + 2(1 + t)v &= 6t(1 + t) \end{aligned}$$

The left-hand side is now the derivative of $(1+t)^2v$, so

$$\frac{d}{dt}((1+t)^2v) = 6t(1+t)$$

Integrating with respect to t ,

$$\begin{aligned}(1+t)^2v &= \int 6t(1+t) \, dt \\ &= \int (6t + 6t^2) \, dt \\ &= 3t^2 + 2t^3 + C\end{aligned}$$

Use the initial condition $v = -5$ when $t = 0$:

$$\begin{aligned}(1+0)^2(-5) &= 3(0)^2 + 2(0)^3 + C \\ -5 &= C\end{aligned}$$

So

$$(1+t)^2v = 3t^2 + 2t^3 - 5$$

and therefore

$$v = \frac{2t^3 + 3t^2 - 5}{(1+t)^2}$$

Factorising the numerator,

$$2t^3 + 3t^2 - 5 = (t-1)(2t^2 + 5t + 5)$$

so

$$v = \frac{(t-1)(2t^2 + 5t + 5)}{(1+t)^2}$$

The particle is stationary when $v = 0$, so we solve

$$\frac{(t-1)(2t^2 + 5t + 5)}{(1+t)^2} = 0$$

For $t \geq 0$, we have $(1+t)^2 > 0$, and also

$$2t^2 + 5t + 5 > 0$$

so the only way $v = 0$ is when

$$t - 1 = 0$$

Hence

$$t = 1$$

Therefore, the first time at which B is stationary is 1 s.

3. A brief flushing cycle in an industrial pipe is to be modelled. The flow rate of liquid, R litres per second, at time t seconds is measured from the instant the valve is opened. The cycle starts when $t = 0$ and there is initially no flow through the pipe. When $t = 1$, the flow rate is measured as $\pi e^{-1} \sin(1)$ litres per second.

While the cycle is occurring, R is modelled by the differential equation

$$\frac{dR}{dt} + (1 - \cot t)R = \frac{e^{-t} \sin t}{1 + t^2}$$

- (a) By using an integrating factor show that, according to the model, while the cycle is occurring,

$$R = e^{-t} \sin t \left(\tan^{-1} t + \frac{3\pi}{4} \right) \quad [6]$$

The cycle ends when there is again no flow through the pipe.

- (b) Determine the length of time that the cycle lasts according to the model [2]
 (c) Determine, according to the model, the rate of increase of the flow rate at the start of the cycle. Give your answer in exact form [3]

Solution

- (a) For the cycle we consider $0 < t < \pi$, so $\sin t > 0$. The differential equation is

$$\frac{dR}{dt} + (1 - \cot t)R = \frac{e^{-t} \sin t}{1 + t^2}$$

An integrating factor is

$$\begin{aligned} \mu(t) &= e^{\int (1 - \cot t) dt} \\ &= e^{t - \int \cot t dt} \\ &= e^{t - \ln(\sin t)} \\ &= \frac{e^t}{\sin t} \end{aligned}$$

Multiplying the differential equation by $\frac{e^t}{\sin t}$ gives

$$\begin{aligned} \frac{e^t}{\sin t} \frac{dR}{dt} + \frac{e^t}{\sin t} (1 - \cot t)R &= \frac{e^t}{\sin t} \cdot \frac{e^{-t} \sin t}{1 + t^2} \\ &= \frac{1}{1 + t^2} \end{aligned}$$

Now

$$\begin{aligned} \frac{d}{dt} \left(\frac{e^t}{\sin t} \right) &= \frac{e^t}{\sin t} - \frac{e^t \cos t}{\sin^2 t} \\ &= \frac{e^t}{\sin t} \left(1 - \frac{\cos t}{\sin t} \right) \\ &= \frac{e^t}{\sin t} (1 - \cot t) \end{aligned}$$

so the left-hand side is

$$\frac{d}{dt} \left(\frac{e^t}{\sin t} R \right)$$

Hence

$$\frac{d}{dt} \left(\frac{e^t}{\sin t} R \right) = \frac{1}{1+t^2}$$

Integrating:

$$\begin{aligned} \frac{e^t}{\sin t} R &= \int \frac{1}{1+t^2} dt \\ &= \tan^{-1} t + C \end{aligned}$$

Therefore

$$R = e^{-t} \sin t (\tan^{-1} t + C)$$

Use the condition $R(1) = \pi e^{-1} \sin 1$:

$$\begin{aligned} \pi e^{-1} \sin 1 &= e^{-1} \sin 1 (\tan^{-1} 1 + C) \\ &= e^{-1} \sin 1 \left(\frac{\pi}{4} + C \right) \end{aligned}$$

Since $e^{-1} \sin 1 \neq 0$,

$$\begin{aligned} \pi &= \frac{\pi}{4} + C \\ C &= \frac{3\pi}{4} \end{aligned}$$

So

$$R = e^{-t} \sin t \left(\tan^{-1} t + \frac{3\pi}{4} \right)$$

Thus, according to the model,

$$R = e^{-t} \sin t \left(\tan^{-1} t + \frac{3\pi}{4} \right)$$

(b) The cycle ends when there is again no flow, so we set $R = 0$:

$$e^{-t} \sin t \left(\tan^{-1} t + \frac{3\pi}{4} \right) = 0$$

For $t \geq 0$, we have $e^{-t} > 0$ and

$$\tan^{-1} t + \frac{3\pi}{4} > 0$$

so the only way $R = 0$ is if

$$\sin t = 0$$

Hence

$$t = n\pi \quad (n \in \mathbb{Z})$$

and the first time after $t = 0$ is

$$t = \pi$$

Therefore the cycle lasts π seconds.

(c) The rate of increase of the flow rate at the start is $R'(0)$.

From part (a),

$$R(t) = e^{-t} \sin t \left(\tan^{-1} t + \frac{3\pi}{4} \right)$$

Differentiate:

$$R'(t) = e^{-t} \left[\cos t \left(\tan^{-1} t + \frac{3\pi}{4} \right) + \frac{\sin t}{1+t^2} - \sin t \left(\tan^{-1} t + \frac{3\pi}{4} \right) \right]$$

Hence

$$R'(0) = 1 \left[1 \cdot \frac{3\pi}{4} + 0 - 0 \right] = \frac{3\pi}{4}$$

So the rate of increase of the flow rate at the start is

$$\frac{3\pi}{4} \text{ L s}^{-2}$$

4. A mixing tank initially contains 600 litres of pure water.

Brine flows into the tank at a constant rate of 20 litres per hour. Each litre of brine contains 3 grams of dissolved salt.

The well-stirred mixture is pumped out of the tank at a constant rate of 30 litres per hour.

It is assumed that the salt is instantly and uniformly mixed throughout the liquid.

Given that there are x grams of salt in the tank after t hours,

(a) show that, while the tank still contains liquid, the situation can be modelled by

$$\frac{dx}{dt} = 60 - \frac{3x}{60 - t} \quad [4]$$

(b) Hence find the number of grams of salt in the tank after 30 hours [5]

(c) State one limitation of this model and explain briefly how the model could be refined [1]

Solution

(a) Let the volume of liquid in the tank after t hours be V litres.

Since liquid flows in at 20 litres/h and out at 30 litres/h, the net change in volume is

$$V = 600 + 20t - 30t = 600 - 10t = 10(60 - t)$$

This is valid while the tank still contains liquid, so $0 \leq t < 60$.

The rate at which salt enters is

$$20 \times 3 = 60 \text{ g/h}$$

The concentration of salt in the tank at time t is

$$\frac{x}{V} = \frac{x}{600 - 10t} \text{ g/litre}$$

Since the mixture leaves at 30 litres/h, the rate at which salt leaves is

$$30 \left(\frac{x}{600 - 10t} \right) = \frac{30x}{600 - 10t} = \frac{3x}{60 - t} \text{ g/h}$$

Hence,

$$\frac{dx}{dt} = \text{rate in} - \text{rate out}$$

so

$$\frac{dx}{dt} = 60 - \frac{3x}{60 - t}$$

Therefore, while the tank contains liquid,

$$\frac{dx}{dt} = 60 - \frac{3x}{60 - t}$$

(b) We solve

$$\frac{dx}{dt} + \frac{3x}{60 - t} = 60$$

with the initial condition $x(0) = 0$, since the tank initially contains pure water.

The integrating factor is

$$\begin{aligned}\mu(t) &= \exp\left(\int \frac{3}{60-t} dt\right) \\ &= \exp(-3\ln(60-t)) \\ &= (60-t)^{-3}\end{aligned}$$

Multiplying the differential equation by $(60-t)^{-3}$ gives

$$(60-t)^{-3} \frac{dx}{dt} + \frac{3x}{60-t} (60-t)^{-3} = 60(60-t)^{-3}$$

The left-hand side is now the derivative of $x(60-t)^{-3}$, so

$$\frac{d}{dt} [x(60-t)^{-3}] = 60(60-t)^{-3}$$

Integrating both sides with respect to t ,

$$x(60-t)^{-3} = \int 60(60-t)^{-3} dt + C$$

To evaluate the integral, let $u = 60 - t$, so $du = -dt$:

$$\begin{aligned}\int 60(60-t)^{-3} dt &= -60 \int u^{-3} du \\ &= -60 \left(\frac{u^{-2}}{-2} \right) \\ &= 30u^{-2} \\ &= 30(60-t)^{-2}\end{aligned}$$

Therefore,

$$x(60-t)^{-3} = 30(60-t)^{-2} + C$$

Multiplying through by $(60-t)^3$,

$$x = 30(60-t) + C(60-t)^3$$

Use $x(0) = 0$:

$$\begin{aligned}0 &= 30(60) + C(60)^3 \\ 0 &= 1800 + 216000C \\ C &= -\frac{1800}{216000} = -\frac{1}{120}\end{aligned}$$

So

$$x = 30(60-t) - \frac{(60-t)^3}{120}$$

At $t = 30$,

$$\begin{aligned}x(30) &= 30(60-30) - \frac{(60-30)^3}{120} \\ &= 30 \cdot 30 - \frac{30^3}{120} \\ &= 900 - \frac{27000}{120} \\ &= 900 - 225 \\ &= 675\end{aligned}$$

Hence the number of grams of salt in the tank after 30 hours is

$$675 \text{ g}$$

- (c) One limitation is that the model assumes the salt is mixed instantly and uniformly throughout the tank, which is not completely realistic.

A refinement would be to use a model with finite mixing time, for example a compartment model or a model where concentration varies in different parts of the tank.

5. The concentration C , in suitable units, of a pollutant in a reservoir at time t hours after a filtration system is switched on is to be modelled by a differential equation. Initially, the concentration is A , and its long-term value is 0.

- (a) One simple model is to assume that the rate of decrease of concentration is directly proportional to C .

(i) Formulate a differential equation for this model [1]

(ii) Verify that $C = Ae^{-kt}$, where k is a positive constant, satisfies this differential equation, the initial condition and the long-term condition [3]

An alternative model uses the differential equation

$$\frac{dC}{dt} + \frac{2t}{1+t^2}C = R(t)$$

where $R(t)$ is a function of t .

- (b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$1 + t^2 \quad [3]$$

- (c) Suppose that $R(t) = 0$.

(i) Show that

$$C = \frac{A}{1+t^2} \quad [4]$$

(ii) Find the time predicted by this model for the concentration to fall to half its initial value. Give your answer correct to the nearest minute [2]

- (d) Now suppose that

$$R(t) = \frac{te^{-t}}{1+t^2}$$

Show that

$$C = \frac{A+1-(1+t)e^{-t}}{1+t^2} \quad [6]$$

It is found that the initial concentration is 5, and the concentration falls to half this value after 63 minutes.

- (e) Determine which of the models proposed in parts (c) and (d) is more consistent with these data [2]

Solution

- (a) (i) If the rate of decrease of concentration is directly proportional to C , then

$$-\frac{dC}{dt} \propto C$$

so, for some constant $k > 0$,

$$\frac{dC}{dt} = -kC$$

- (ii) Take

$$C = Ae^{-kt}$$

Then

$$\frac{dC}{dt} = \frac{d}{dt}(Ae^{-kt}) = -kAe^{-kt}$$

Since $Ae^{-kt} = C$, this becomes

$$\frac{dC}{dt} = -kC$$

so the differential equation is satisfied.

For the initial condition,

$$C(0) = Ae^{-k \cdot 0} = A$$

so the initial concentration is correct.

For the long-term condition, since $k > 0$,

$$\lim_{t \rightarrow \infty} C = \lim_{t \rightarrow \infty} Ae^{-kt} = A \lim_{t \rightarrow \infty} e^{-kt} = 0$$

Hence $C = Ae^{-kt}$ satisfies the differential equation, the initial condition and the long-term condition.

(b) For

$$\frac{dC}{dt} + \frac{2t}{1+t^2}C = R(t)$$

the integrating factor is

$$\mu(t) = e^{\int \frac{2t}{1+t^2} dt}$$

To evaluate the integral, let

$$u = 1 + t^2 \quad \Rightarrow \quad du = 2t dt$$

Then

$$\begin{aligned} \int \frac{2t}{1+t^2} dt &= \int \frac{1}{u} du \\ &= \ln u \\ &= \ln(1+t^2) \end{aligned}$$

since $1+t^2 > 0$.

Therefore

$$\mu(t) = e^{\ln(1+t^2)} = 1+t^2$$

Hence an integrating factor is $1+t^2$.

(c) (i) If $R(t) = 0$, the differential equation becomes

$$\frac{dC}{dt} + \frac{2t}{1+t^2}C = 0$$

Multiply through by the integrating factor $1+t^2$:

$$(1+t^2)\frac{dC}{dt} + 2tC = 0$$

The left-hand side is the derivative of $(1+t^2)C$, so

$$\frac{d}{dt}((1+t^2)C) = 0$$

Integrating gives

$$(1+t^2)C = K$$

where K is a constant.

Use the initial condition $C(0) = A$:

$$(1+0^2)A = K$$

so $K = A$. Therefore

$$C = \frac{A}{1+t^2}$$

(ii) Half the initial concentration is $\frac{A}{2}$, so set

$$\frac{A}{1+t^2} = \frac{A}{2}$$

Cancelling A ,

$$\frac{1}{1+t^2} = \frac{1}{2}$$

so

$$1+t^2 = 2$$

$$t^2 = 1$$

Since $t \geq 0$,

$$t = 1 \text{ hour}$$

Therefore the time is 60 minutes.

(d) If

$$R(t) = \frac{te^{-t}}{1+t^2}$$

then the differential equation is

$$\frac{dC}{dt} + \frac{2t}{1+t^2}C = \frac{te^{-t}}{1+t^2}$$

Multiply through by the integrating factor $1+t^2$:

$$(1+t^2)\frac{dC}{dt} + 2tC = te^{-t}$$

Hence

$$\frac{d}{dt}((1+t^2)C) = te^{-t}$$

Integrating,

$$(1+t^2)C = \int te^{-t} dt$$

Evaluate the integral by parts. Let

$$u = t, \quad \frac{dv}{dt} = e^{-t}$$

Then

$$\frac{du}{dt} = 1, \quad v = -e^{-t}$$

So

$$\begin{aligned} \int te^{-t} dt &= uv - \int v \frac{du}{dt} dt \\ &= t(-e^{-t}) - \int (-e^{-t}) dt \\ &= -te^{-t} + \int e^{-t} dt \\ &= -te^{-t} - e^{-t} + K \\ &= -(1+t)e^{-t} + K \end{aligned}$$

Therefore

$$(1+t^2)C = -(1+t)e^{-t} + K$$

Use $C(0) = A$:

$$\begin{aligned} (1+0^2)A &= -(1+0)e^0 + K \\ A &= -1 + K \end{aligned}$$

so

$$K = A + 1$$

Hence

$$C = \frac{A + 1 - (1 + t)e^{-t}}{1 + t^2}$$

Also, using $\lim_{t \rightarrow \infty} (1 + t)e^{-t} = 0$,

$$\lim_{t \rightarrow \infty} C = \lim_{t \rightarrow \infty} \frac{A + 1 - (1 + t)e^{-t}}{1 + t^2} = 0$$

so this model also has the correct long-term value.

(e) From part (c), model (c) predicts the concentration falls to half its initial value after 60 minutes.

For model (d), with $A = 5$,

$$C = \frac{6 - (1 + t)e^{-t}}{1 + t^2}$$

Half the initial concentration is 2.5, so solve

$$\frac{6 - (1 + t)e^{-t}}{1 + t^2} = 2.5$$

Using a calculator gives

$$t \approx 1.055 \text{ hours}$$

which is

$$1.055 \times 60 \approx 63.3 \text{ minutes}$$

This is much closer to the observed 63 minutes than the 60 minutes from model (c). Therefore model (d) is more consistent with the data.

6. A processing tank initially contains 15 litres of water in which 6 mg of dye is dissolved.

A dye solution enters the tank at a constant rate of 200 ml per minute. The concentration of dye in this incoming solution, at time t minutes after the flow begins, is modelled by $(4 + e^{-0.1t})$ mg per litre.

At the same time, pure water enters through a second pipe at a constant rate of 100 ml per minute. The liquid in the tank is continually mixed and leaves the tank at a constant rate of 300 ml per minute.

Let m be the amount of dye, in milligrams, in the tank at time t minutes after the flow begins.

- (a) (i) Write down the volume of liquid in the tank at time t
 (ii) Hence show that m satisfies the differential equation

$$\frac{dm}{dt} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50} \quad [4]$$

- (b) By solving the differential equation, find an expression for m in terms of t [8]

After 30 minutes, the concentration of dye in the tank was measured as 1.52 mg per litre.

- (c) Assess the model in light of this information, giving a reason for your answer [2]

Solution

- (a) (i) Convert the flow rates to litres per minute:

$$200 \text{ ml min}^{-1} = 0.2 \text{ L min}^{-1}, \quad 100 \text{ ml min}^{-1} = 0.1 \text{ L min}^{-1}, \quad 300 \text{ ml min}^{-1} = 0.3 \text{ L min}^{-1}$$

So the net rate of change of volume is

$$0.2 + 0.1 - 0.3 = 0 \text{ L min}^{-1}$$

Hence the volume stays constant at its initial value of 15 litres:

$$V(t) = 15$$

So the volume of liquid in the tank at time t is 15 litres.

- (ii) The rate at which dye enters through the first pipe is

$$0.2(4 + e^{-0.1t}) \text{ mg min}^{-1}$$

so

$$\text{rate in} = 0.8 + 0.2e^{-0.1t}$$

The second pipe contains pure water, so it contributes no dye.

Since the tank is well mixed and has volume 15 L, the concentration of dye in the tank is

$$\frac{m}{15} \text{ mg L}^{-1}$$

Therefore the rate at which dye leaves the tank is

$$0.3 \times \frac{m}{15} = \frac{0.3m}{15} = \frac{m}{50} \text{ mg min}^{-1}$$

Hence

$$\frac{dm}{dt} = \text{rate in} - \text{rate out}$$

so

$$\frac{dm}{dt} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50}$$

as required.

(b) Rearranging the differential equation into linear form gives

$$\frac{dm}{dt} + \frac{1}{50}m = 0.8 + 0.2e^{-0.1t}$$

The integrating factor is

$$\exp\left(\int \frac{1}{50} dt\right) = e^{t/50}$$

Multiply the differential equation by $e^{t/50}$:

$$\begin{aligned} e^{t/50} \frac{dm}{dt} + \frac{1}{50} e^{t/50} m &= 0.8e^{t/50} + 0.2e^{-0.1t} e^{t/50} \\ &= 0.8e^{t/50} + 0.2e^{-0.08t} \end{aligned}$$

The left-hand side is

$$\frac{d}{dt}(me^{t/50})$$

so

$$\frac{d}{dt}(me^{t/50}) = 0.8e^{t/50} + 0.2e^{-0.08t}$$

Integrating both sides with respect to t :

$$\begin{aligned} me^{t/50} &= \int 0.8e^{t/50} dt + \int 0.2e^{-0.08t} dt + C \\ &= 0.8(50)e^{t/50} + 0.2\left(\frac{1}{-0.08}\right)e^{-0.08t} + C \\ &= 40e^{t/50} - 2.5e^{-0.08t} + C \end{aligned}$$

Divide through by $e^{t/50}$:

$$\begin{aligned} m &= 40 - 2.5e^{-0.08t}e^{-t/50} + Ce^{-t/50} \\ &= 40 - 2.5e^{-0.1t} + Ce^{-0.02t} \end{aligned}$$

Initially there are 6 mg of dye in the tank, so $m(0) = 6$. Substituting $t = 0$ gives

$$\begin{aligned} 6 &= 40 - 2.5 + C \\ 6 &= 37.5 + C \\ C &= -31.5 \end{aligned}$$

Therefore

$$m = 40 - 2.5e^{-0.1t} - 31.5e^{-0.02t}$$

(c) The model predicts that at $t = 30$ minutes,

$$m(30) = 40 - 2.5e^{-3} - 31.5e^{-0.6}$$

so the concentration is

$$\frac{m(30)}{15} = \frac{40 - 2.5e^{-3} - 31.5e^{-0.6}}{15} \approx 1.51 \text{ mg L}^{-1}$$

The measured concentration was 1.52 mg L^{-1} , which is very close to the model prediction. Therefore the model appears to be reasonable.

7. A particle B , of mass 2 kg, moves in a straight line and has velocity $v \text{ m s}^{-1}$.

At time t seconds, where $t \geq 0$, a variable force of $-2t(v+3) \text{ N}$ acts on B . There are no other forces acting on B . Initially, when $t = 0$, B has velocity 5 m s^{-1} .

The motion of B can be modelled by the differential equation

$$\frac{dv}{dt} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t .

- (a) Find the functions $P(t)$ and $Q(t)$ [2]

- (b) Using an integrating factor, determine the first time at which B is stationary according to the model [8]

Solution

- (a) By Newton's second law, resultant force = mass $\times$ acceleration, so

$$2 \frac{dv}{dt} = -2t(v+3)$$

Expanding and dividing by 2 gives

$$\frac{dv}{dt} = -tv - 3t$$

Hence

$$\frac{dv}{dt} + tv = -3t$$

Therefore, $P(t) = t$ and $Q(t) = -3t$.

- (b) From part (a), the differential equation is

$$\frac{dv}{dt} + tv = -3t$$

The integrating factor is

$$e^{\int t \, dt} = e^{t^2/2}$$

Multiplying the differential equation by $e^{t^2/2}$ gives

$$e^{t^2/2} \frac{dv}{dt} + tve^{t^2/2} = -3te^{t^2/2}$$

The left-hand side is

$$\frac{d}{dt} (ve^{t^2/2})$$

so

$$\frac{d}{dt} (ve^{t^2/2}) = -3te^{t^2/2}$$

Integrating with respect to t ,

$$\begin{aligned} ve^{t^2/2} &= \int -3te^{t^2/2} \, dt \\ &= -3e^{t^2/2} + C \end{aligned}$$

Therefore

$$v = -3 + Ce^{-t^2/2}$$

Using the initial condition $v = 5$ when $t = 0$,

$$5 = -3 + Ce^0$$

$$5 = -3 + C$$

$$C = 8$$

Hence

$$v = -3 + 8e^{-t^2/2}$$

For B to be stationary, $v = 0$. So

$$0 = -3 + 8e^{-t^2/2}$$

$$8e^{-t^2/2} = 3$$

$$e^{-t^2/2} = \frac{3}{8}$$

Taking natural logarithms,

$$-\frac{t^2}{2} = \ln\left(\frac{3}{8}\right)$$

$$\frac{t^2}{2} = \ln\left(\frac{8}{3}\right)$$

$$t^2 = 2\ln\left(\frac{8}{3}\right)$$

Since $t \geq 0$, the first time is

$$t = \sqrt{2\ln\left(\frac{8}{3}\right)}$$

Numerically,

$$t \approx 1.40 \text{ s}$$

Therefore, the first time at which B is stationary is $\sqrt{2\ln\left(\frac{8}{3}\right)}$ s, approximately 1.40 s.

8. A small rescue boat moves in a straight line on calm water. It starts from rest, is driven by its engine for 5 seconds and then, for the next 5 seconds, the engine is cut and a drogue is deployed. The boat comes to rest at the end of the 10 seconds. The total mass of the boat and crew is m kg, and at time t seconds, for $0 \leq t \leq 10$, the boat's velocity is v m s⁻¹.

A resistance to motion, modelled by a force of magnitude $0.2mv$ N, acts on the boat throughout the 10 seconds.

- (a) Explain why modelling the resistance to motion in this way is likely to be more realistic than assuming this force is constant [1]

- (b) During the drogue phase of the motion, for $5 \leq t \leq 10$, the drogue exerts an additional constant resistance force of magnitude m N and the engine provides no driving force. Show that, for $5 \leq t \leq 10$,

$$\frac{dv}{dt} + 0.2v = -1 \quad [1]$$

- (c) (i) Solve the differential equation in part (b) [4]
 (ii) Hence find the velocity of the boat when $t = 5$ [1]

- (d) During the powered phase ($0 \leq t \leq 5$), the propeller provides a driving force of magnitude directly proportional to t^2 . Show that, for $0 \leq t \leq 5$,

$$\frac{dv}{dt} + 0.2v = \lambda t^2 \quad [1]$$

where λ is a positive constant

- (e) (i) Show by integration that, for $0 \leq t \leq 5$,

$$v = 5\lambda(t^2 - 10t + 50 - 50e^{-t/5}) \quad [6]$$

- (ii) Hence find λ [2]

- (f) Find the total distance, to the nearest metre, travelled by the boat during the motion [6]

Solution

- (a) A resistive force depending on v is more realistic because water resistance increases as speed increases, and when $v = 0$ this model gives zero resistance. A constant resistive force would not reflect either of these features.

- (b) During $5 \leq t \leq 10$, the only horizontal forces are the two resistances:

$$0.2mv \quad \text{and} \quad m$$

both acting opposite to the motion. Taking the direction of motion as positive and using Newton's second law,

$$m \frac{dv}{dt} = -0.2mv - m$$

Dividing by m gives

$$\frac{dv}{dt} + 0.2v = -1$$

as required.

- (c) (i) For $5 \leq t \leq 10$, we solve

$$\frac{dv}{dt} + \frac{1}{5}v = -1$$

The integrating factor is

$$e^{\int \frac{1}{5} dt} = e^{t/5}$$

Multiplying the differential equation by $e^{t/5}$,

$$\begin{aligned} e^{t/5} \frac{dv}{dt} + \frac{1}{5} e^{t/5} v &= -e^{t/5} \\ \frac{d}{dt} (ve^{t/5}) &= -e^{t/5} \end{aligned}$$

Integrating,

$$\begin{aligned} ve^{t/5} &= \int -e^{t/5} dt \\ &= -5e^{t/5} + C \end{aligned}$$

Hence

$$v = -5 + Ce^{-t/5}$$

The boat comes to rest at $t = 10$, so $v = 0$ when $t = 10$:

$$\begin{aligned} 0 &= -5 + Ce^{-2} \\ Ce^{-2} &= 5 \\ C &= 5e^2 \end{aligned}$$

Therefore

$$v = 5e^2 e^{-t/5} - 5 = 5(e^{2-t/5} - 1)$$

So the solution is $v = 5(e^{2-t/5} - 1)$ for $5 \leq t \leq 10$.

(ii) Substituting $t = 5$ into the result from part (i),

$$\begin{aligned} v(5) &= 5(e^{2-1} - 1) \\ &= 5(e - 1) \end{aligned}$$

Hence the velocity when $t = 5$ is $5(e - 1) \text{ m s}^{-1}$.

- (d) Let the driving force during $0 \leq t \leq 5$ be λmt^2 , where $\lambda > 0$ is a constant. Then Newton's second law gives

$$m \frac{dv}{dt} = \lambda mt^2 - 0.2mv$$

Dividing by m ,

$$\frac{dv}{dt} + 0.2v = \lambda t^2$$

as required.

- (e) (i) We solve

$$\frac{dv}{dt} + \frac{1}{5}v = \lambda t^2$$

The integrating factor is again

$$e^{t/5}$$

Multiplying through by $e^{t/5}$,

$$\begin{aligned} e^{t/5} \frac{dv}{dt} + \frac{1}{5} e^{t/5} v &= \lambda t^2 e^{t/5} \\ \frac{d}{dt} (ve^{t/5}) &= \lambda t^2 e^{t/5} \end{aligned}$$

So

$$ve^{t/5} = \lambda \int t^2 e^{t/5} dt + C$$

Let

$$I = \int t^2 e^{t/5} dt$$

Integrating by parts with $u = t^2$ and $\frac{dw}{dt} = e^{t/5}$,

$$\begin{aligned} I &= 5t^2 e^{t/5} - \int 10te^{t/5} dt \\ &= 5t^2 e^{t/5} - 10 \int te^{t/5} dt \end{aligned}$$

Now let

$$J = \int te^{t/5} dt$$

Again integrating by parts,

$$\begin{aligned} J &= 5te^{t/5} - \int 5e^{t/5} dt \\ &= 5te^{t/5} - 25e^{t/5} \end{aligned}$$

Hence

$$\begin{aligned} I &= 5t^2 e^{t/5} - 10 \left(5te^{t/5} - 25e^{t/5} \right) \\ &= 5t^2 e^{t/5} - 50te^{t/5} + 250e^{t/5} \\ &= 5e^{t/5} (t^2 - 10t + 50) \end{aligned}$$

Therefore

$$\begin{aligned} ve^{t/5} &= \lambda \left[5e^{t/5} (t^2 - 10t + 50) \right] + C \\ v &= 5\lambda (t^2 - 10t + 50) + Ce^{-t/5} \end{aligned}$$

The boat starts from rest, so $v = 0$ when $t = 0$:

$$\begin{aligned} 0 &= 5\lambda(50) + C \\ C &= -250\lambda \end{aligned}$$

Hence

$$\begin{aligned} v &= 5\lambda (t^2 - 10t + 50) - 250\lambda e^{-t/5} \\ &= 5\lambda \left(t^2 - 10t + 50 - 50e^{-t/5} \right) \end{aligned}$$

Thus, for $0 \leq t \leq 5$,

$$v = 5\lambda \left(t^2 - 10t + 50 - 50e^{-t/5} \right)$$

- (ii) At $t = 5$, the expressions for v from parts (c) and (e)(i) must be equal.
From part (e)(i),

$$\begin{aligned} v(5) &= 5\lambda (25 - 50 + 50 - 50e^{-1}) \\ &= 5\lambda \left(25 - \frac{50}{e} \right) \end{aligned}$$

From part (c)(ii),

$$v(5) = 5(e - 1)$$

Equating these,

$$\begin{aligned} 5\lambda \left(25 - \frac{50}{e} \right) &= 5(e - 1) \\ \lambda \left(25 - \frac{50}{e} \right) &= e - 1 \\ \lambda \left(\frac{25e - 50}{e} \right) &= e - 1 \\ \lambda \left(\frac{25(e - 2)}{e} \right) &= e - 1 \end{aligned}$$

So

$$\lambda = \frac{e(e-1)}{25(e-2)}$$

Therefore

$$\lambda = \frac{e(e-1)}{25(e-2)} \approx 0.260$$

(f) Since the boat moves forwards throughout the motion, the total distance is

$$s = \int_0^{10} v \, dt = \int_0^5 v \, dt + \int_5^{10} v \, dt$$

For the powered phase,

$$\begin{aligned} s_1 &= \int_0^5 5\lambda (t^2 - 10t + 50 - 50e^{-t/5}) \, dt \\ &= 5\lambda \left[\frac{t^3}{3} - 5t^2 + 50t + 250e^{-t/5} \right]_0^5 \\ &= 5\lambda \left(\frac{125}{3} - 125 + 250 + \frac{250}{e} - 250 \right) \\ &= 5\lambda \left(\frac{125}{3} - 125 + \frac{250}{e} \right) \\ &= 5\lambda \left(-\frac{250}{3} + \frac{250}{e} \right) \\ &= 1250\lambda \left(\frac{3-e}{3e} \right) \end{aligned}$$

Substituting $\lambda = \frac{e(e-1)}{25(e-2)}$,

$$\begin{aligned} s_1 &= 1250 \left(\frac{e(e-1)}{25(e-2)} \right) \left(\frac{3-e}{3e} \right) \\ &= \frac{50(e-1)(3-e)}{3(e-2)} \end{aligned}$$

So

$$s_1 \approx 11.2 \text{ m}$$

For the drogue phase,

$$\begin{aligned} s_2 &= \int_5^{10} 5(e^{2-t/5} - 1) \, dt \\ &= 5 \left[\int_5^{10} e^{2-t/5} \, dt - \int_5^{10} 1 \, dt \right] \\ &= 5 \left[-5e^{2-t/5} - t \right]_5^{10} \\ &= 5((-5 - 10) - (-5e - 5)) \\ &= 25(e - 2) \end{aligned}$$

So

$$s_2 \approx 18.0 \text{ m}$$

Therefore the total distance is

$$\begin{aligned} s &= s_1 + s_2 \\ &= \frac{50(e-1)(3-e)}{3(e-2)} + 25(e-2) \\ &\approx 11.2 + 18.0 \\ &\approx 29.2 \end{aligned}$$

Hence the total distance travelled, to the nearest metre, is 29 m.

9. A winch raises a lift cage vertically. The cage starts from rest and comes to rest again 8 seconds later. The combined mass of the cage and its load is m kg, and at time t seconds, for $0 \leq t \leq 8$, the cage has upward velocity v m s⁻¹.
A resistance to the motion, modelled by a force of magnitude $0.25mv$ N, acts on the cage throughout the 8 seconds.
In this question, take $g = 9.8$ m s⁻².

- (a) Explain why modelling the resistance to the motion in this way is likely to be more realistic than assuming this force is constant [1]
- (b) During the final stage of the motion, for $4 \leq t \leq 8$, the winch provides a constant upward tension of magnitude $0.75mg$ N. Show that, for $4 \leq t \leq 8$,

$$\frac{dv}{dt} + 0.25v = -\frac{1}{4}g \quad [1]$$

- (c) (i) Solve the differential equation in part (b) [5]
(ii) Hence find the velocity of the cage when $t = 4$ [1]
- (d) During the first stage of the motion, for $0 \leq t \leq 4$, the winch provides an upward tension of magnitude $m(g + \lambda t)$ N, where λ is a positive constant. Show that, for $0 \leq t \leq 4$,

$$\frac{dv}{dt} + 0.25v = \lambda t \quad [1]$$

- (e) (i) Show by integration that, for $0 \leq t \leq 4$,

$$v = 4\lambda(t - 4 + 4e^{-t/4}) \quad [5]$$

- (ii) Hence find λ [2]

- (f) Find the total distance, to the nearest metre, through which the cage rises during the motion [6]

Solution

- (a) A resistive force usually depends on the speed of the moving object. When the cage moves faster, the resistance is greater, so modelling it as proportional to v is more realistic than taking it to be constant.
- (b) For $4 \leq t \leq 8$, taking upwards as positive:

$$\text{tension} = 0.75mg, \quad \text{weight} = mg, \quad \text{resistance} = 0.25mv$$

Using Newton's second law,

$$\begin{aligned} m \frac{dv}{dt} &= 0.75mg - mg - 0.25mv \\ &= -0.25mg - 0.25mv \end{aligned}$$

Dividing by m ,

$$\frac{dv}{dt} = -0.25g - 0.25v$$

Hence

$$\frac{dv}{dt} + 0.25v = -\frac{1}{4}g$$

(c) (i) For $4 \leq t \leq 8$, solve

$$\frac{dv}{dt} + 0.25v = -\frac{1}{4}g$$

The integrating factor is

$$e^{\int 0.25 dt} = e^{t/4}$$

Multiplying the differential equation by $e^{t/4}$,

$$e^{t/4} \frac{dv}{dt} + 0.25e^{t/4}v = -\frac{1}{4}ge^{t/4}$$

The left-hand side is

$$\frac{d}{dt} (ve^{t/4})$$

so

$$\frac{d}{dt} (ve^{t/4}) = -\frac{1}{4}ge^{t/4}$$

Integrating,

$$\begin{aligned} ve^{t/4} &= \int -\frac{1}{4}ge^{t/4} dt \\ &= -ge^{t/4} + C \end{aligned}$$

Hence

$$v = Ce^{-t/4} - g$$

The cage comes to rest at $t = 8$, so $v(8) = 0$. Therefore

$$0 = Ce^{-8/4} - g$$

$$0 = Ce^{-2} - g$$

$$C = ge^2$$

Substituting for C ,

$$\begin{aligned} v &= ge^2e^{-t/4} - g \\ &= g(e^{2-t/4} - 1) \end{aligned}$$

So, for $4 \leq t \leq 8$,

$$v = g(e^{2-t/4} - 1)$$

(ii) Using the result from part (i),

$$\begin{aligned} v(4) &= g(e^{2-4/4} - 1) \\ &= g(e^1 - 1) \\ &= g(e - 1) \end{aligned}$$

With $g = 9.8$,

$$v(4) \approx 9.8(e - 1) \approx 16.8 \text{ m s}^{-1}$$

So the velocity when $t = 4$ is $g(e - 1) \text{ m s}^{-1} \approx 16.8 \text{ m s}^{-1}$.

(d) For $0 \leq t \leq 4$, the upward tension is $m(g + \lambda t)$.

Using Newton's second law with upwards positive,

$$\begin{aligned} m \frac{dv}{dt} &= m(g + \lambda t) - mg - 0.25mv \\ &= m\lambda t - 0.25mv \end{aligned}$$

Dividing by m ,

$$\frac{dv}{dt} + 0.25v = \lambda t$$

(e) (i) For $0 \leq t \leq 4$, solve

$$\frac{dv}{dt} + 0.25v = \lambda t$$

The integrating factor is

$$e^{\int 0.25 \, dt} = e^{t/4}$$

Multiplying through by $e^{t/4}$,

$$e^{t/4} \frac{dv}{dt} + 0.25e^{t/4}v = \lambda te^{t/4}$$

Hence

$$\frac{d}{dt} (ve^{t/4}) = \lambda te^{t/4}$$

Integrating,

$$ve^{t/4} = \lambda \int te^{t/4} \, dt + C$$

Evaluate the integral by parts. Let

$$u = t, \quad \frac{dw}{dt} = e^{t/4}$$

so that

$$\frac{du}{dt} = 1, \quad w = 4e^{t/4}$$

Then

$$\begin{aligned} \int te^{t/4} \, dt &= 4te^{t/4} - \int 4e^{t/4} \, dt \\ &= 4te^{t/4} - 16e^{t/4} \end{aligned}$$

Therefore

$$ve^{t/4} = \lambda (4te^{t/4} - 16e^{t/4}) + C$$

Dividing by $e^{t/4}$,

$$v = 4\lambda(t - 4) + Ce^{-t/4}$$

The cage starts from rest, so $v(0) = 0$. Hence

$$0 = 4\lambda(0 - 4) + C$$

$$0 = -16\lambda + C$$

$$C = 16\lambda$$

Substituting back,

$$\begin{aligned} v &= 4\lambda(t - 4) + 16\lambda e^{-t/4} \\ &= 4\lambda(t - 4 + 4e^{-t/4}) \end{aligned}$$

So, for $0 \leq t \leq 4$,

$$v = 4\lambda(t - 4 + 4e^{-t/4})$$

(ii) The velocity is continuous at $t = 4$, so the expressions from parts (c)(ii) and (e)(i) must be equal. From part (e)(i),

$$\begin{aligned} v(4) &= 4\lambda(4 - 4 + 4e^{-1}) \\ &= \frac{16\lambda}{e} \end{aligned}$$

From part (c)(ii),

$$v(4) = g(e - 1)$$

Therefore

$$\begin{aligned}\frac{16\lambda}{e} &= g(e - 1) \\ \lambda &= \frac{ge(e - 1)}{16} \\ &= \frac{g(e^2 - e)}{16}\end{aligned}$$

With $g = 9.8$,

$$\lambda \approx \frac{9.8 \times e(e - 1)}{16} \approx 2.86$$

So

$$\lambda = \frac{ge(e - 1)}{16} \approx 2.86$$

(f) The total distance risen is

$$s = \int_0^4 v \, dt + \int_4^8 v \, dt$$

Using the expressions found for v ,

$$\begin{aligned}s_1 &= \int_0^4 4\lambda (t - 4 + 4e^{-t/4}) \, dt \\ &= 4\lambda \left(\int_0^4 t \, dt - \int_0^4 4 \, dt + \int_0^4 4e^{-t/4} \, dt \right) \\ &= 4\lambda \left(\left[\frac{t^2}{2} \right]_0^4 - [4t]_0^4 + [-16e^{-t/4}]_0^4 \right) \\ &= 4\lambda \left(8 - 16 + \left(-\frac{16}{e} + 16 \right) \right) \\ &= 4\lambda \left(8 - \frac{16}{e} \right) \\ &= \left(32 - \frac{64}{e} \right) \lambda\end{aligned}$$

For the second stage,

$$\begin{aligned}s_2 &= \int_4^8 g(e^{2-t/4} - 1) \, dt \\ &= g \left(\int_4^8 e^{2-t/4} \, dt - \int_4^8 1 \, dt \right) \\ &= g \left([-4e^{2-t/4}]_4^8 - [t]_4^8 \right) \\ &= g((-4 + 4e) - 4) \\ &= 4g(e - 2)\end{aligned}$$

So

$$\begin{aligned}s &= s_1 + s_2 \\ &= \left(32 - \frac{64}{e} \right) \lambda + 4g(e - 2)\end{aligned}$$

Substitute $\lambda = \frac{ge(e-1)}{16}$:

$$\begin{aligned}s &= \left(32 - \frac{64}{e}\right) \frac{ge(e-1)}{16} + 4g(e-2) \\&= \left(2 - \frac{4}{e}\right) ge(e-1) + 4g(e-2) \\&= (2e-4)g(e-1) + 4g(e-2) \\&= 2g(e-2)(e-1) + 4g(e-2) \\&= 2g(e-2)(e+1) \\&= 2g(e^2 - e - 2)\end{aligned}$$

Now use $g = 9.8$:

$$\begin{aligned}s &\approx 2(9.8)(e^2 - e - 2) \\&\approx 52.3 \text{ m}\end{aligned}$$

Therefore the total distance risen, to the nearest metre, is 52 m.

10. A short filling pulse in a buffer tank is to be modelled. The volume of liquid in the tank at time t seconds is V litres. The pulse starts when $t = 0$, and initially the tank is empty. After 1 second the volume is measured to be 4 litres. While the pulse is occurring, V is modelled by the differential equation

$$(3t - t^2) \frac{dV}{dt} = \frac{(3t - t^2)^2}{1 + (t - 1)^2} + (3 - 2t)V$$

- (a) By using an integrating factor show that, according to the model, while the pulse is occurring,

$$V = (3t - t^2) (\tan^{-1}(t - 1) + 2) \quad [6]$$

The pulse ends when the tank is again empty.

- (b) Determine the length of time that the pulse lasts according to the model [2]
- (c) Determine, according to the model, the rate of increase of the volume at the start of the pulse. Give your answer in an exact form [3]

Solution

- (a) First rearrange the differential equation into linear form:

$$\frac{dV}{dt} - \frac{3 - 2t}{3t - t^2} V = \frac{3t - t^2}{1 + (t - 1)^2}$$

The integrating factor is

$$\mu(t) = e^{\int -\frac{3-2t}{3t-t^2} dt}$$

Since

$$\frac{d}{dt}(3t - t^2) = 3 - 2t$$

we have

$$\int -\frac{3 - 2t}{3t - t^2} dt = -\ln(3t - t^2)$$

Hence

$$\mu(t) = e^{-\ln(3t - t^2)} = \frac{1}{3t - t^2}$$

Multiplying the differential equation by this integrating factor gives

$$\frac{1}{3t - t^2} \frac{dV}{dt} - \frac{3 - 2t}{(3t - t^2)^2} V = \frac{1}{1 + (t - 1)^2}$$

The left-hand side is

$$\frac{d}{dt} \left(\frac{V}{3t - t^2} \right)$$

so

$$\frac{d}{dt} \left(\frac{V}{3t - t^2} \right) = \frac{1}{1 + (t - 1)^2}$$

Integrating with respect to t ,

$$\begin{aligned} \frac{V}{3t - t^2} &= \int \frac{1}{1 + (t - 1)^2} dt \\ &= \tan^{-1}(t - 1) + C \end{aligned}$$

Use the given condition $V = 4$ when $t = 1$:

$$\begin{aligned}\frac{4}{3(1) - 1^2} &= \tan^{-1}(1 - 1) + C \\ \frac{4}{2} &= 0 + C \\ C &= 2\end{aligned}$$

Therefore

$$\frac{V}{3t - t^2} = \tan^{-1}(t - 1) + 2$$

and so

$$V = (3t - t^2) (\tan^{-1}(t - 1) + 2)$$

This is the required result.

(b) The pulse ends when the tank is empty again, so set $V = 0$:

$$(3t - t^2) (\tan^{-1}(t - 1) + 2) = 0$$

Now

$$-\frac{\pi}{2} < \tan^{-1}(t - 1) < \frac{\pi}{2}$$

so

$$2 - \frac{\pi}{2} < \tan^{-1}(t - 1) + 2 < 2 + \frac{\pi}{2}$$

Since $2 - \frac{\pi}{2} > 0$, the factor $\tan^{-1}(t - 1) + 2$ is never zero. Hence

$$3t - t^2 = 0$$

so

$$\begin{aligned}t(3 - t) &= 0 \\ t &= 0 \text{ or } t = 3\end{aligned}$$

The pulse starts at $t = 0$, so it ends when $t = 3$. Therefore the pulse lasts 3 seconds.

(c) Differentiate the expression for V :

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt} [(3t - t^2) (\tan^{-1}(t - 1) + 2)] \\ &= (3 - 2t) (\tan^{-1}(t - 1) + 2) + (3t - t^2) \left(\frac{1}{1 + (t - 1)^2} \right)\end{aligned}$$

At the start of the pulse, $t = 0$:

$$\begin{aligned}\left. \frac{dV}{dt} \right|_{t=0} &= 3 (\tan^{-1}(-1) + 2) + 0 \\ &= 3 \left(-\frac{\pi}{4} + 2 \right) \\ &= 6 - \frac{3\pi}{4}\end{aligned}$$

Therefore the rate of increase of volume at the start is $6 - \frac{3\pi}{4} \text{ L s}^{-1}$.

11. A horticultural mixing tank initially contains 600 litres of pure water.

A fertiliser solution flows into the tank at a constant rate of 6 litres per minute. The incoming solution contains 2 grams of fertiliser in every litre of solution.

At the same time, liquid is pumped out of the well-stirred tank at a constant rate of 9 litres per minute.

It is assumed that the fertiliser instantly dissolves uniformly throughout the tank.

Given that there are x grams of fertiliser in the tank after t minutes,

- (a) Show that, while the tank still contains liquid, the situation can be modelled by the differential equation

$$\frac{dx}{dt} = 12 - \frac{3x}{200 - t} \quad [4]$$

- (b) Hence determine after how many minutes the concentration of fertiliser in the tank first reaches 1.5 grams per litre. [5]

- (c) Explain briefly one way in which the model could be refined. [1]

Solution

- (a) Let V be the volume of liquid in the tank after t minutes.

Since liquid enters at 6 L/min and leaves at 9 L/min, the net change in volume is

$$\begin{aligned} V(t) &= 600 + (6 - 9)t \\ &= 600 - 3t \\ &= 3(200 - t) \end{aligned}$$

This is valid while the tank still contains liquid, so

$$600 - 3t > 0 \quad \Rightarrow \quad t < 200$$

The fertiliser enters at

$$6 \times 2 = 12 \text{ g/min}$$

The concentration in the tank at time t is

$$\frac{x}{V} = \frac{x}{600 - 3t} \text{ g/L}$$

So the fertiliser leaves at the rate

$$\begin{aligned} 9 \times \frac{x}{600 - 3t} &= \frac{9x}{600 - 3t} \\ &= \frac{9x}{3(200 - t)} \\ &= \frac{3x}{200 - t} \end{aligned}$$

Hence,

$$\begin{aligned} \frac{dx}{dt} &= \text{rate in} - \text{rate out} \\ &= 12 - \frac{3x}{200 - t} \end{aligned}$$

So the model is

$$\frac{dx}{dt} = 12 - \frac{3x}{200 - t}, \quad t < 200$$

(b) From part (a),

$$\frac{dx}{dt} + \frac{3}{200-t}x = 12$$

This is a linear differential equation. Its integrating factor is

$$\mu(t) = \exp\left(\int \frac{3}{200-t} dt\right) = \exp(-3\ln(200-t)) = (200-t)^{-3}$$

Multiplying the differential equation by $(200-t)^{-3}$ gives

$$\begin{aligned}(200-t)^{-3} \frac{dx}{dt} + \frac{3x}{200-t}(200-t)^{-3} &= 12(200-t)^{-3} \\ \frac{d}{dt}(x(200-t)^{-3}) &= 12(200-t)^{-3}\end{aligned}$$

Integrating,

$$\begin{aligned}x(200-t)^{-3} &= \int 12(200-t)^{-3} dt \\ &= 6(200-t)^{-2} + C\end{aligned}$$

Therefore,

$$\begin{aligned}x &= (6(200-t)^{-2} + C)(200-t)^3 \\ &= 6(200-t) + C(200-t)^3\end{aligned}$$

Initially the tank contains pure water, so $x = 0$ when $t = 0$. Substituting $t = 0$ gives

$$\begin{aligned}0 &= 6(200) + C(200)^3 \\ 0 &= 1200 + 8,000,000C \\ C &= -\frac{1200}{8,000,000} \\ &= -\frac{3}{20000}\end{aligned}$$

Hence

$$x = 6(200-t) - \frac{3}{20000}(200-t)^3$$

The volume at time t is

$$V = 3(200-t)$$

So the concentration is

$$\begin{aligned}\frac{x}{V} &= \frac{6(200-t) - \frac{3}{20000}(200-t)^3}{3(200-t)} \\ &= 2 - \frac{(200-t)^2}{20000}\end{aligned}$$

We want this concentration to be 1.5 g/L, so

$$\begin{aligned}2 - \frac{(200-t)^2}{20000} &= 1.5 \\ \frac{(200-t)^2}{20000} &= 0.5 \\ (200-t)^2 &= 10000\end{aligned}$$

Since $t < 200$, we must have $200-t > 0$. Therefore

$$200-t = 100$$

So

$$t = 100$$

Therefore the concentration first reaches 1.5 g/L after 100 minutes.

- (c) One refinement would be to remove the assumption of perfect instantaneous mixing, and allow the fertiliser concentration to vary in different parts of the tank.

So a more realistic model could account for imperfect mixing.

- 12.** A mixing tank initially contains 2 litres of solution in which 4 grams of dye are dissolved.
 A dye solution containing 10 grams of dye per litre is pumped into the tank at a constant rate of r litres per minute and mixes instantly and uniformly with the liquid already in the tank.
 At the same time, liquid is removed from the tank at a constant rate of s litres per minute.
 After t minutes, the tank contains m grams of dye.

- (a) (i) Show that the concentration of dye in the tank after t minutes is

$$\frac{m}{2 + (r - s)t}$$

grams per litre.

[2]

- (ii) Hence show that

$$\frac{dm}{dt} + \frac{sm}{2 + (r - s)t} = 10r$$

[2]

- (b) First, consider the case where $s = r$.

- (i) Solve the differential equation to find m in terms of r and t .

[4]

- (ii) Given that after 2 minutes the concentration of dye in the tank is 8 grams per litre, find r .

[2]

- (c) Now consider the case where $s = 2r$.

- (i) Explain why the differential equation in part (a)(ii) is no longer valid for $t \geq \frac{2}{r}$.

[1]

- (ii) Find the maximum mass of dye in the tank.

[9]

Solution

- (a) (i) Initially the tank contains 2 litres.

In t minutes, liquid enters at r litres per minute, so volume added is rt litres.

In the same time, liquid leaves at s litres per minute, so volume removed is st litres.

Hence the volume in the tank after t minutes is

$$2 + rt - st = 2 + (r - s)t$$

Since the mass of dye is m grams, the concentration is

$$\frac{m}{2 + (r - s)t}$$

So the concentration of dye after t minutes is $\frac{m}{2 + (r - s)t}$ grams per litre.

- (ii) The rate at which dye enters the tank is

$$10 \times r = 10r$$

grams per minute.

From part (i), the concentration in the tank is $\frac{m}{2 + (r - s)t}$ grams per litre, so the rate at which dye leaves is

$$s \cdot \frac{m}{2 + (r - s)t}$$

grams per minute.

Therefore

$$\frac{dm}{dt} = \text{rate in} - \text{rate out}$$

so

$$\frac{dm}{dt} = 10r - \frac{sm}{2 + (r - s)t}$$

Rearranging gives

$$\frac{dm}{dt} + \frac{sm}{2 + (r - s)t} = 10r$$

(b) (i) If $s = r$, then part (a)(ii) becomes

$$\frac{dm}{dt} + \frac{rm}{2 + (r - r)t} = 10r$$

so

$$\frac{dm}{dt} + \frac{r}{2}m = 10r$$

Also, initially there are 4 grams of dye, so

$$m(0) = 4$$

This is a linear differential equation. Its integrating factor is

$$\exp\left(\int \frac{r}{2} dt\right) = e^{rt/2}$$

Multiplying the differential equation by $e^{rt/2}$ gives

$$e^{rt/2} \frac{dm}{dt} + \frac{r}{2} e^{rt/2} m = 10r e^{rt/2}$$

The left-hand side is the derivative of $me^{rt/2}$, so

$$\frac{d}{dt} \left(me^{rt/2} \right) = 10r e^{rt/2}$$

Integrating,

$$\begin{aligned} me^{rt/2} &= \int 10r e^{rt/2} dt \\ &= 10r \cdot \frac{2}{r} e^{rt/2} + C \\ &= 20e^{rt/2} + C \end{aligned}$$

Therefore

$$m = 20 + C e^{-rt/2}$$

Using $m(0) = 4$,

$$\begin{aligned} 4 &= 20 + C e^0 \\ 4 &= 20 + C \\ C &= -16 \end{aligned}$$

Hence

$$m = 20 - 16e^{-rt/2}$$

(ii) When $s = r$, the volume is constant and equal to

$$2 + (r - r)t = 2$$

So the concentration is

$$\frac{m}{2} = \frac{20 - 16e^{-rt/2}}{2} = 10 - 8e^{-rt/2}$$

After 2 minutes, the concentration is 8 grams per litre, so

$$8 = 10 - 8e^{-r}$$

Solving,

$$\begin{aligned}8e^{-r} &= 2 \\e^{-r} &= \frac{1}{4} \\-r &= \ln\left(\frac{1}{4}\right) \\r &= \ln 4\end{aligned}$$

Therefore

$$r = \ln 4 \approx 1.39$$

So the required inflow rate is $r = \ln 4$ litres per minute.

- (c) (i) If $s = 2r$, then the volume after t minutes is

$$2 + (r - 2r)t = 2 - rt$$

At

$$t = \frac{2}{r}$$

this volume is 0, so the tank is empty.

For $t \geq \frac{2}{r}$, the denominator in the differential equation is 0 or negative, so the model is no longer physically valid.

Hence the differential equation is not valid for $t \geq \frac{2}{r}$ because the tank is empty at $t = \frac{2}{r}$.

- (ii) With $s = 2r$, the differential equation from part (a)(ii) is

$$\frac{dm}{dt} + \frac{2rm}{2 + (r - 2r)t} = 10r$$

so

$$\frac{dm}{dt} + \frac{2r}{2 - rt}m = 10r$$

with

$$m(0) = 4$$

We solve this for $0 \leq t < \frac{2}{r}$.

The integrating factor is

$$\begin{aligned}\exp\left(\int \frac{2r}{2 - rt} dt\right) &= \exp(-2 \ln(2 - rt)) \\&= (2 - rt)^{-2}\end{aligned}$$

Multiplying through by $(2 - rt)^{-2}$ gives

$$(2 - rt)^{-2} \frac{dm}{dt} + \frac{2r}{2 - rt} (2 - rt)^{-2} m = 10r(2 - rt)^{-2}$$

The left-hand side is

$$\frac{d}{dt} (m(2 - rt)^{-2})$$

So

$$\frac{d}{dt} (m(2 - rt)^{-2}) = 10r(2 - rt)^{-2}$$

Integrating,

$$\begin{aligned}m(2 - rt)^{-2} &= \int 10r(2 - rt)^{-2} dt \\&= \frac{10}{2 - rt} + C\end{aligned}$$

Therefore

$$m = 10(2 - rt) + C(2 - rt)^2$$

Using $m(0) = 4$,

$$4 = 10(2) + C(2)^2$$

$$4 = 20 + 4C$$

$$4C = -16$$

$$C = -4$$

Hence

$$m = 10(2 - rt) - 4(2 - rt)^2$$

Expanding,

$$\begin{aligned} m &= 20 - 10rt - 4(4 - 4rt + r^2t^2) \\ &= 20 - 10rt - 16 + 16rt - 4r^2t^2 \\ &= 4 + 6rt - 4r^2t^2 \end{aligned}$$

This is a quadratic in t with negative coefficient of t^2 , so it has a maximum at its stationary point.

Differentiate:

$$\frac{dm}{dt} = 6r - 8r^2t$$

Set this equal to 0:

$$6r - 8r^2t = 0$$

$$6r = 8r^2t$$

$$6 = 8rt$$

$$t = \frac{3}{4r}$$

Since $\frac{3}{4r} < \frac{2}{r}$, this occurs before the tank becomes empty, so it is valid.

Now substitute $t = \frac{3}{4r}$ into $m = 4 + 6rt - 4r^2t^2$:

$$\begin{aligned} m_{\max} &= 4 + 6r \left(\frac{3}{4r} \right) - 4r^2 \left(\frac{3}{4r} \right)^2 \\ &= 4 + \frac{18}{4} - 4r^2 \cdot \frac{9}{16r^2} \\ &= 4 + \frac{18}{4} - \frac{36}{16} \\ &= 4 + \frac{9}{2} - \frac{9}{4} \\ &= \frac{16}{4} + \frac{18}{4} - \frac{9}{4} \\ &= \frac{25}{4} \end{aligned}$$

Therefore the maximum mass of dye in the tank is

$$\frac{25}{4} \text{ g}$$

- 13.** The concentration C , in suitable units, of a drug in a patient's bloodstream at time t hours after an intravenous infusion begins is to be modelled. Initially, the concentration is zero, and its long-term value is M .

(a) One simple model assumes that the rate of change of concentration is directly proportional to $M - C$.

(i) Formulate a differential equation for this model. [1]

(ii) Verify that $C = M(1 - e^{-kt})$, where k is a positive constant, satisfies this differential equation, the initial condition and the long-term condition. [3]

An alternative model uses, for $t > 0$, the differential equation

$$\frac{dC}{dt} - \frac{t+2}{t(1+t+t^2)} C = S(t),$$

where $S(t)$ is a function of t .

(b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$\frac{1+t+t^2}{t^2} \quad [8]$$

(c) Suppose that $S(t) = 0$.

(i) Show that

$$C = \frac{Mt^2}{1+t+t^2} \quad [4]$$

(ii) Find the time predicted by this model for the concentration to reach half its long-term value. Give your answer correct to the nearest minute. [2]

(d) Now suppose that

$$S(t) = \frac{3t^2 e^{-t}}{1+t+t^2}$$

Show that

$$C = \frac{Mt^2 - 3t^2 e^{-t}}{1+t+t^2} \quad [5]$$

(e) It is found that the long-term concentration is 10, and C reaches half this value after 105 minutes. Determine which of the models proposed in parts (c) and (d) is more consistent with this data. [2]

Solution

(a) (i) Since the rate of change is directly proportional to $M - C$,

$$\frac{dC}{dt} = k(M - C), \quad k > 0$$

(ii) Take

$$C = M(1 - e^{-kt})$$

Differentiate:

$$\begin{aligned}\frac{dC}{dt} &= \frac{d}{dt} [M(1 - e^{-kt})] \\ &= M(0 - (-k)e^{-kt}) \\ &= Mke^{-kt}\end{aligned}$$

Also,

$$\begin{aligned}M - C &= M - M(1 - e^{-kt}) \\ &= M - M + Me^{-kt} \\ &= Me^{-kt}\end{aligned}$$

Hence,

$$\frac{dC}{dt} = Mke^{-kt} = k(Me^{-kt}) = k(M - C)$$

so the differential equation is satisfied.

For the initial condition,

$$C(0) = M(1 - e^0) = M(1 - 1) = 0$$

For the long-term condition, since $k > 0$, $e^{-kt} \rightarrow 0$ as $t \rightarrow \infty$. Therefore

$$\lim_{t \rightarrow \infty} C = \lim_{t \rightarrow \infty} M(1 - e^{-kt}) = M(1 - 0) = M$$

So $C = M(1 - e^{-kt})$ satisfies the differential equation, the initial condition, and the long-term condition.

(b) Write the equation in the linear form

$$\frac{dC}{dt} + P(t)C = S(t)$$

where

$$P(t) = -\frac{t+2}{t(1+t+t^2)}$$

To find the integrating factor, first decompose $P(t)$ into partial fractions:

$$-\frac{t+2}{t(1+t+t^2)} = \frac{A}{t} + \frac{Bt+D}{1+t+t^2}$$

Multiplying through by $t(1+t+t^2)$ gives

$$\begin{aligned}-(t+2) &= A(1+t+t^2) + t(Bt+D) \\ &= A + At + At^2 + Bt^2 + Dt \\ &= A + (A+D)t + (A+B)t^2\end{aligned}$$

Comparing coefficients with $-2 - t + 0t^2$:

$$\begin{aligned}A &= -2 \\ A + D &= -1 \\ A + B &= 0\end{aligned}$$

So

$$A = -2, \quad D = 1, \quad B = 2$$

Therefore

$$P(t) = -\frac{2}{t} + \frac{2t+1}{1+t+t^2}$$

The integrating factor is

$$\mu(t) = e^{\int P(t) dt}$$

Since $t > 0$,

$$\begin{aligned}\int P(t) \, dt &= \int \left(-\frac{2}{t} + \frac{2t+1}{1+t+t^2} \right) \, dt \\ &= -2 \ln t + \ln(1+t+t^2)\end{aligned}$$

Hence

$$\begin{aligned}\mu(t) &= e^{-2 \ln t + \ln(1+t+t^2)} \\ &= e^{\ln(1+t+t^2)} e^{-2 \ln t} \\ &= (1+t+t^2)t^{-2} \\ &= \frac{1+t+t^2}{t^2}\end{aligned}$$

So the integrating factor is $\frac{1+t+t^2}{t^2}$.

(c) (i) When $S(t) = 0$, the differential equation becomes

$$\frac{dC}{dt} - \frac{t+2}{t(1+t+t^2)}C = 0$$

Using the integrating factor

$$\mu(t) = \frac{1+t+t^2}{t^2}$$

gives

$$\frac{d}{dt}(\mu C) = 0$$

So

$$\mu C = A$$

where A is a constant. Thus

$$\begin{aligned}C &= \frac{A}{\mu} \\ &= \frac{A}{(1+t+t^2)/t^2} \\ &= \frac{At^2}{1+t+t^2}\end{aligned}$$

Now use the long-term value:

$$\begin{aligned}\lim_{t \rightarrow \infty} C &= \lim_{t \rightarrow \infty} \frac{At^2}{1+t+t^2} \\ &= \lim_{t \rightarrow \infty} \frac{A}{t^{-2} + t^{-1} + 1} \\ &= A\end{aligned}$$

Since the long-term concentration is M , we must have $A = M$.

Therefore

$$C = \frac{Mt^2}{1+t+t^2}$$

(ii) Half the long-term value is $\frac{M}{2}$. Set

$$\frac{Mt^2}{1+t+t^2} = \frac{M}{2}$$

Cancelling M and solving:

$$\begin{aligned}\frac{t^2}{1+t+t^2} &= \frac{1}{2} \\ 2t^2 &= 1+t+t^2 \\ t^2 - t - 1 &= 0\end{aligned}$$

Hence

$$t = \frac{1 \pm \sqrt{5}}{2}$$

Since $t > 0$,

$$t = \frac{1 + \sqrt{5}}{2} \text{ hours} \approx 1.618 \text{ hours}$$

Converting to minutes:

$$1.618 \times 60 \approx 97.1$$

So the concentration reaches half its long-term value after 97 minutes, to the nearest minute.

(d) Using the integrating factor from part (b),

$$\mu(t) = \frac{1 + t + t^2}{t^2}$$

and

$$S(t) = \frac{3t^2 e^{-t}}{1 + t + t^2}$$

Then

$$\begin{aligned} \mu(t)S(t) &= \frac{1 + t + t^2}{t^2} \cdot \frac{3t^2 e^{-t}}{1 + t + t^2} \\ &= 3e^{-t} \end{aligned}$$

So the differential equation becomes

$$\frac{d}{dt}(\mu C) = 3e^{-t}$$

Integrating:

$$\begin{aligned} \mu C &= \int 3e^{-t} dt \\ &= -3e^{-t} + A \end{aligned}$$

Hence

$$\begin{aligned} C &= \frac{-3e^{-t} + A}{\mu} \\ &= \frac{t^2(-3e^{-t} + A)}{1 + t + t^2} \\ &= \frac{At^2 - 3t^2 e^{-t}}{1 + t + t^2} \end{aligned}$$

Now use the long-term value. As $t \rightarrow \infty$,

$$\frac{At^2}{1 + t + t^2} \rightarrow A$$

and, using the fact that $\lim_{t \rightarrow \infty} t^2 e^{-t} = 0$,

$$\frac{3t^2 e^{-t}}{1 + t + t^2} \rightarrow 0$$

Therefore

$$\lim_{t \rightarrow \infty} C = A$$

Since the long-term concentration is M , $A = M$.

So

$$C = \frac{Mt^2 - 3t^2 e^{-t}}{1 + t + t^2}$$

as required.

- (e) Here $M = 10$, so half the long-term concentration is 5, reached after 105 minutes = 1.75 hours.
From part (c), model (c) predicts the half-value time to be 97 minutes.
For model (d), at $t = 1.75$,

$$\begin{aligned} C &= \frac{10t^2 - 3t^2e^{-t}}{1 + t + t^2} \\ &= \frac{10(1.75)^2 - 3(1.75)^2e^{-1.75}}{1 + 1.75 + (1.75)^2} \\ &\approx \frac{30.625 - 1.597}{5.8125} \\ &\approx 4.99 \end{aligned}$$

This is essentially 5.

Therefore model (d) is more consistent with the data.

14. A drinks manufacturer uses a mixing tank that initially contains 4 litres of water. A flavouring liquid, which is 60% fruit concentrate by volume, is pumped into the tank at a constant rate of p litres per minute. At the same time, the contents of the tank are thoroughly mixed and drawn off at a constant rate of q litres per minute.

After t minutes, let x litres be the amount of fruit concentrate in the tank. You may assume that mixing is instantaneous and uniform.

- (a) (i) Show that the proportion of fruit concentrate in the tank after t minutes is

$$\frac{x}{4 + (p - q)t} \quad [2]$$

- (ii) Hence show that

$$\frac{dx}{dt} + \frac{qx}{4 + (p - q)t} = \frac{3p}{5} \quad [2]$$

- (b) Now consider the case where $q = p$.

- (i) Solve the differential equation to find x in terms of p and t . [4]

- (ii) Given that after 4 minutes the liquid in the tank is 30% fruit concentrate, find the value of p . [2]

- (c) Now consider the case where $q = 2p$.

- (i) Explain why the differential equation in part (a)(ii) is no longer valid for $t \geq \frac{4}{p}$. [1]

- (ii) Find the maximum amount of fruit concentrate in the tank before the tank empties. [9]

Solution

- (a) (i) After t minutes, the volume in the tank is

$$4 + pt - qt = 4 + (p - q)t$$

Since there are x litres of fruit concentrate in the tank, the proportion of fruit concentrate is

$$\frac{x}{4 + (p - q)t}$$

- (ii) The flavouring liquid entering the tank is 60% concentrate, so the rate at which concentrate enters is

$$0.6p = \frac{3p}{5}$$

From part (i), the proportion of concentrate in the tank is $\frac{x}{4 + (p - q)t}$, so the rate at which concentrate leaves is

$$q \left(\frac{x}{4 + (p - q)t} \right) = \frac{qx}{4 + (p - q)t}$$

Therefore

$$\begin{aligned} \frac{dx}{dt} &= \text{rate in} - \text{rate out} \\ &= \frac{3p}{5} - \frac{qx}{4 + (p - q)t} \end{aligned}$$

Rearranging,

$$\frac{dx}{dt} + \frac{qx}{4 + (p - q)t} = \frac{3p}{5}$$

Hence the differential equation is shown.

(b) (i) If $q = p$, then the differential equation from part (a)(ii) becomes

$$\frac{dx}{dt} + \frac{px}{4 + (p - p)t} = \frac{3p}{5}$$

so

$$\frac{dx}{dt} + \frac{p}{4}x = \frac{3p}{5}$$

Initially the tank contains only water, so

$$x(0) = 0$$

The integrating factor is

$$\mu(t) = e^{\int \frac{p}{4} dt} = e^{pt/4}$$

Multiplying the differential equation by $e^{pt/4}$ gives

$$\begin{aligned} e^{pt/4} \frac{dx}{dt} + \frac{p}{4} e^{pt/4} x &= \frac{3p}{5} e^{pt/4} \\ \frac{d}{dt} (x e^{pt/4}) &= \frac{3p}{5} e^{pt/4} \end{aligned}$$

Integrating,

$$\begin{aligned} x e^{pt/4} &= \int \frac{3p}{5} e^{pt/4} dt \\ &= \frac{3p}{5} \cdot \frac{4}{p} e^{pt/4} + C \\ &= \frac{12}{5} e^{pt/4} + C \end{aligned}$$

Hence

$$x = \frac{12}{5} + C e^{-pt/4}$$

Using $x(0) = 0$,

$$0 = \frac{12}{5} + C$$

so

$$C = -\frac{12}{5}$$

Therefore

$$x = \frac{12}{5} (1 - e^{-pt/4})$$

(ii) When $q = p$, the total volume stays constant at 4 litres.

After 4 minutes, the liquid is 30% concentrate, so the amount of concentrate is

$$x = 0.3 \times 4 = \frac{6}{5}$$

Substitute $t = 4$ into the formula from part (i):

$$\frac{12}{5} (1 - e^{-p}) = \frac{6}{5}$$

Multiply through by 5:

$$12(1 - e^{-p}) = 6$$

Hence

$$\begin{aligned}1 - e^{-p} &= \frac{1}{2} \\ e^{-p} &= \frac{1}{2}\end{aligned}$$

Taking natural logarithms,

$$-p = \ln\left(\frac{1}{2}\right) = -\ln 2$$

so

$$p = \ln 2$$

Therefore $p = \ln 2 \approx 0.693$ litres min^{-1} .

- (c) (i) If $q = 2p$, then the volume after t minutes is

$$4 + (p - 2p)t = 4 - pt$$

At

$$t = \frac{4}{p}$$

this volume is 0, so the tank is empty. For $t \geq \frac{4}{p}$, the denominator $4 - pt$ is zero or negative, which is not physically meaningful.

Therefore the differential equation is only valid for $0 \leq t < \frac{4}{p}$.

- (ii) With $q = 2p$, the differential equation becomes

$$\frac{dx}{dt} + \frac{2px}{4 - pt} = \frac{3p}{5}$$

for $0 \leq t < \frac{4}{p}$, with

$$x(0) = 0$$

The integrating factor is

$$\mu(t) = e^{\int \frac{2p}{4-pt} dt}$$

Now let $u = 4 - pt$, so $du = -p dt$. Then

$$\begin{aligned}\int \frac{2p}{4 - pt} dt &= -2 \int \frac{1}{u} du \\ &= -2 \ln u \\ &= -2 \ln(4 - pt)\end{aligned}$$

Hence

$$\mu(t) = e^{-2 \ln(4-pt)} = (4 - pt)^{-2}$$

Multiply the differential equation by $(4 - pt)^{-2}$:

$$(4 - pt)^{-2} \frac{dx}{dt} + \frac{2p}{4 - pt} (4 - pt)^{-2} x = \frac{3p}{5} (4 - pt)^{-2}$$

The left-hand side is

$$\frac{d}{dt} (x(4 - pt)^{-2})$$

so

$$\frac{d}{dt} (x(4 - pt)^{-2}) = \frac{3p}{5} (4 - pt)^{-2}$$

Integrating,

$$x(4 - pt)^{-2} = \int \frac{3p}{5} (4 - pt)^{-2} dt$$

Again let $u = 4 - pt$, so $du = -p dt$. Then

$$\begin{aligned}\int \frac{3p}{5}(4 - pt)^{-2} dt &= -\frac{3}{5} \int u^{-2} du \\ &= -\frac{3}{5} \left(\frac{u^{-1}}{-1} \right) + C \\ &= \frac{3}{5} u^{-1} + C \\ &= \frac{3}{5(4 - pt)} + C\end{aligned}$$

Therefore

$$x(4 - pt)^{-2} = \frac{3}{5(4 - pt)} + C$$

Multiply by $(4 - pt)^2$:

$$x = \frac{3}{5}(4 - pt) + C(4 - pt)^2$$

Use $x(0) = 0$:

$$\begin{aligned}0 &= \frac{3}{5}(4) + C(4)^2 \\ 0 &= \frac{12}{5} + 16C\end{aligned}$$

so

$$C = -\frac{3}{20}$$

Hence

$$\begin{aligned}x &= \frac{3}{5}(4 - pt) - \frac{3}{20}(4 - pt)^2 \\ &= \frac{12}{5} - \frac{3pt}{5} - \frac{3}{20}(16 - 8pt + p^2t^2) \\ &= \frac{12}{5} - \frac{3pt}{5} - \frac{12}{5} + \frac{6pt}{5} - \frac{3p^2t^2}{20} \\ &= \frac{3pt}{5} - \frac{3p^2t^2}{20}\end{aligned}$$

So

$$x = \frac{3pt}{5} - \frac{3p^2t^2}{20} = \frac{3}{20}pt(4 - pt)$$

To find the maximum value, differentiate:

$$\frac{dx}{dt} = \frac{3p}{5} - \frac{3p^2t}{10}$$

Set this equal to 0:

$$\begin{aligned}\frac{3p}{5} - \frac{3p^2t}{10} &= 0 \\ \frac{3p}{10}(2 - pt) &= 0\end{aligned}$$

Since $p > 0$, this gives

$$2 - pt = 0$$

so

$$t = \frac{2}{p}$$

This lies in the interval $0 \leq t < \frac{4}{p}$, and since the quadratic has negative coefficient of t^2 , this gives the maximum.

Now substitute $t = \frac{2}{p}$:

$$\begin{aligned}x_{\max} &= \frac{3p}{5} \left(\frac{2}{p} \right) - \frac{3p^2}{20} \left(\frac{2}{p} \right)^2 \\&= \frac{6}{5} - \frac{3}{20} \cdot 4 \\&= \frac{6}{5} - \frac{12}{20} \\&= \frac{6}{5} - \frac{3}{5} \\&= \frac{3}{5}\end{aligned}$$

Therefore the maximum amount of fruit concentrate is $\frac{3}{5}$ litres, occurring at $t = \frac{2}{p}$ minutes.